

Evaluation-Guided Domain Randomization for Robust Controller Learning

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Abstract

Controllers learned in simulation can perform unevenly across plant parameters and initial conditions. We propose to adapt training distributions using estimates of where a controller meets a performance requirement. Each round evaluates a frozen controller, targets training near its performance boundary, and evaluates the update on fresh data. Linear systems and statistical certification provide the course foundation. CartPole experiments and analysis of improvement and regression form the main research, supporting an ambitious goal: conditions for preserving or expanding acceptable regions under restricted controller updates.

1. Motivation

Learning from simulation helps control nonlinear systems when direct design is difficult. Domain randomization trains across varying conditions, but its distribution affects where a controller performs well. We investigate whether evaluation can guide this distribution to improve coverage of a specified operating domain, connecting data selection for controller learning to closed-loop robustness analysis.

2. Problem Formulation

The proposed method connects three feedback loops:

- **Control:** the policy observes the system and applies actions, producing a closed-loop trajectory in each environment.
- **Evaluation:** adaptively selected rollouts of a frozen policy estimate its performance level set and inform training-condition selection.
- **Improvement:** the estimate guides the next training distribution; the updated policy is frozen and evaluated again.

At outer round k , freeze policy π_k ; $t = 0, \dots, H - 1$ indexes rollout time. A condition $z = (\theta, s) \in \mathcal{Z}$ specifies plant parameters θ and initial-state distribution $x_0 \sim \nu_s$; remaining randomness has a declared conditional law. For failure tolerance α , let $p_k(z)$ be the task-failure probability and $S_k = \{z : p_k(z) \leq \alpha\}$ the acceptable region. Under a fixed reference probability measure μ , coverage is $V_k = \mu(S_k)$. Gains $g_k = \mu(S_{k+1} \setminus S_k)$ and losses $\ell_k = \mu(S_k \setminus S_{k+1})$ satisfy $V_{k+1} - V_k = g_k - \ell_k$. Training distribution q_k adapts to evaluation evidence; μ stays fixed. Certificates require direct rollout evidence and apply only to the declared evaluation grid.

RQ1 (empirical). Does targeting training near the estimated reliability boundary improve acceptable coverage compared with uniform and intermediate-difficulty sampling, under matched budgets? **RQ2 (theoretical).** Which assumptions on a parameterized system family, controller updates, and performance margins bound regression or permit acceptable-region expansion?

Test problems. A small linear family $x_{t+1} = A(\theta)x_t + B(\theta)u_t$, with $u_t = -Kx_t$, supports interpretable analysis. CartPole tests nonlinear policy learning over initial angle and pole half-length. With its angle termination cutoff relaxed to 90° , recovery requires survival for 500 steps and final-100-step RMS angle at most 5° . Other termination rules are retained. Recovery certificates concern finite-horizon performance, not asymptotic stability.

3. Goals and Approach

Stage 1: Course foundation. We will learn a linear controller under a fixed domain-randomization distribution and derive a finite-horizon quadratic-cost sensitivity bound. Writing $J_{H,K}(\theta, x) = x^\top P_H(K, \theta)x$, for $\|x\|_2 \leq r$ an update satisfies $|J_{H,K'} - J_{H,K}| \leq r^2 \|P_H(K', \theta) - P_H(K, \theta)\|_2 =: b(\theta)$. Thus $J_{H,K} \leq h - b(\theta)$ ensures preservation below cost threshold h . In parallel, we will implement level-set estimation and time-uniform confidence bounds: certify a grid condition acceptable when its failure-probability upper bound is at most α . Fresh conditionally independent episodes and error allocation across conditions, methods, and rounds yield simultaneous validity with error budget δ . Certified acceptable coverage at a fixed rollout budget assesses this evaluator.

Stage 2: Main research. We will use the resulting evaluation evidence to adapt $q_k = (1 - \lambda)q_{\text{base}} + \lambda q_{\text{target},k}$. Matched PPO branches from a common checkpoint compare uniform targets, conditions near $\hat{p}_k = \alpha$, and intermediate difficulty scored by $\hat{p}_k(1 - \hat{p}_k)$. The latter favors mixed outcomes rather than threshold proximity. We hold the background mixture, training objective, optimizer, and budgets fixed and collect fresh on-policy training trajectories. Fresh reference evaluations assess gains, losses, and net coverage with uncertainty, seed variation, and total cost, separating performance improvement from certificate growth. The linear family supports analysis of these tradeoffs, including counterexamples to unconditional improvement.

Stage 3: Ambitious theoretical outcome. Building on the sensitivity result, we will seek restricted updates whose performance drift and margins ensure preservation or net expansion. Both cost and failure-probability level sets remain candidate targets; transferring cost bounds to probabilities requires additional assumptions. Whether repeated rounds are required remains open. The foundation and main study remain deliverables without an expansion theorem.

4. Related Work

Control and synthesis. Fujinami et al. (2025) analyze policy gradient for domain-randomized LQR under bounded heterogeneity; Fujinami et al. (2026) extend this to history-dependent controllers with a memory-expansion curriculum. Their average-cost analysis motivates our linear study. **Evaluation.** Gotovos et al. (2013) study active level-set estimation; Letham et al. (2022) address Bernoulli observations; and Howard et al. (2021) develop confidence sequences. **Improvement.** Reverse curricula (Florensa et al., 2017), prioritized level replay (Jiang et al., 2021), and sampling for learnability (Rutherford et al., 2024) motivate our training-condition comparisons. Lyapunov-based safe-region expansion (Berkenkamp et al., 2017) concerns stability, distinct from our recovery criterion.

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